

Name: _____

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Homework

#04: Shift, Rotate and Logical Operations

Shift and Rotate Operations Codes

What is the hexadecimal result of the following shift operation (hexadecimal value in R0)? And what are the values of condition flags? Assume all the flags are cleared before the operations. (Show your work.)

1.	<pre>MOV32 R1, #0xA0AEFEF ASRS R0, R1, #3</pre> ANS: _____. N: _____; Z: _____; C: _____.
2.	<pre>MOV32 R1, #0x1234ABFF MOV R2, #0x6 LSR R0, R1, R2</pre> ANS: _____. N: _____; Z: _____; C: _____.
3.	<pre>MOV32 R1, #0x1234ABCD MOV R2, #0x4 LSLS R0, R1, R2</pre> ANS: _____. N: _____; Z: _____; C: _____.

4.	<pre>MOV32 R1, #0xDD22EE77 MOV R2, #0x3 RORS R0, R1, R2</pre> ANS: _____. N: _____; Z: _____; C: _____.

Logical Operations

5. What is the hexadecimal result of the following **OR** operation? (Show your work.)

#0xC8 | #0xEF = _____

6. What is the hexadecimal result of the following **AND** operation? (Show your work.)

#0x51 & #0x9D = _____

7. What is the hexadecimal result of the following **EOR** (exclusive OR) operation? (Show your work.)

#0x3B ^ #0x67 = _____

Logical Masks

8. Assume that the following 32-bit binary value is stored in **R0**. You want to use a logical operation to guarantee that **bits 13, 11, 5, and 1** in this register are cleared and then store the result back in **R0**. Remembering that the AND operation can be used to clear individual bits, what mask value would you load into **R3** so that the assembly instruction given would accomplish this (i.e., **bits 13, 11, 5, & 1 clear**)? Give your answer in **hexadecimal**.

0001.0101.1001.0100.11**11**.**1**100.10**00**.11**10** (original data)
_____._____._____._____._____._____._____._____. (binary mask)

AND R0, R0, R3

ANS: _____ (mask in hexadecimal)

9. Assume that the following 32-bit binary value is stored in **R0**. You want to use a logical operation to guarantee that **bits 19, 18, 17, and 16** in this register are set high and then store the result back in **R0**. Remembering that the OR operation can be used to set individual bits, what mask value would you load into **R3** so that the assembly instruction given would accomplish this (i.e., **bits 19, 18, 17, & 16 set**)? Give your answer in **hexadecimal**.

0001.0101.1001.**0100**.1111.1100.1000.1110 (original data)
_____._____._____._____._____._____._____._____. (binary mask)

ORR R0, R0, R3

ANS: _____ (mask in hexadecimal)

10. Assume that the following 32-bit binary value is stored in **R0**. You want to use a logical operation to guarantee that **bits 8-15** in this register are toggled (flipped) and then store the result back in **R0**. Remembering that the XOR operation can be used to toggle individual bits, what mask value would you load into **R3** so that the assembly instruction given would accomplish this (i.e., **bits 8-15 flipped**)? Give your answer in **hexadecimal**.

0001.0101.1001.0100.**1111.1100**.1000.1110 (original data)
_____._____._____._____._____._____._____._____. (binary mask)

EOR R0, R0, R3

ANS: _____ (mask in hexadecimal)